Ritesh Somashekar

AI & ML Engineer

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PROFESSIONAL SUMMARY

- **AI & Machine Learning Engineer** with **5+ years** progressing from **data engineering** to **machine learning** and **Generative AI**, building **production RAG systems** and **LLM applications**; published at **ACM WSDM '26** and **IEEE PST 2026**.
- Skilled in **Llama 3, LlamaIndex, ChromaDB, pgvector, PyTorch, Airflow, Spark,** and **Snowflake** on **AWS** and **Azure**, with **MLOps** via **MLflow, Docker, Kubernetes, FastAPI,** and **CI/CD** for **HIPAA-compliant** data.
- Demonstrated impact, including **RAG recall@10 raised from 61% to 88%**, a **0.81 protocol F1 with a 3.0% hallucination rate**, **99.9% pipeline reliability**, and **60% faster model training**, delivering **end-to-end AI solutions** from research to production.

TECHNICAL SKILLS

Programming & Scripting: Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow), SQL, R, Scala, Java, Bash

RAG & Generative AI: RAG, Chunking, Embeddings (all-mpnet-base-v2), Metadata Filtering, Top-K Retrieval, Context Assembly, LlamaIndex, ChromaDB, pgvector, FAISS, Llama 3/3.1, Ollama, Prompt Engineering, Agentic AI, LoRA/PEFT, QAT

LLM Evaluation & Safety: RAGAS, Recall@K, Faithfulness, Hallucination Detection, Guardrails, Ontology-Based Validation, NLI-Based Validation, Semantic Similarity Scoring, Ablation Studies, Protocol F1, Confidence Scoring

Machine Learning & NLP: Transformers, BERT, BioBERT, Hugging Face, XGBoost, LightGBM, Graph Analytics, Ensemble Models, Time Series Forecasting, Anomaly Detection, Feature Engineering, SHAP

MLOps & ML Infrastructure: MLflow, Docker, Kubernetes, Jenkins, CI/CD, FastAPI, Git, DVC, Model Deployment

Data Engineering & Big Data: Apache Airflow, Apache Spark, PySpark, Kafka, Databricks, Snowflake, Delta Lake, Hadoop, Hive, Sqoop, Informatica PowerCenter, Precisely Data360, SQL Server, MySQL, MongoDB

Cloud & Visualization: AWS (S3, Lambda, Athena, Glue, SageMaker, SQS), Azure (Data Services, Stream Analytics), Tableau, Power BI (DAX), Excel (VBA)

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.

PROFESSIONAL EXPERIENCE

George Mason University | *AI & ML Research Associate* **Fairfax, VA | August 2024 - Present**

- Co-developed **SHIELD** (**ACM WSDM '26**), a conversational AI platform for 9-1-1 call-taker training with **Fairfax County DPSC**, using **LLM agents** to generate scenarios from CAD data and deliver **real-time metacognitive nudges**.
- Co-developed **MATE-911** (**IEEE PST 2026**), a multimodal **RAG** system grounding **Llama 3.1 8B** in emergency protocols, reaching **0.81 protocol F1**, a **3.0% hallucination rate**, and a **97.6% pass rate** on local hardware.
- **Led the RAG knowledge base and data pipeline for both platforms**: OCR parsing, abbreviation expansion, completeness checks, and segmentation of **160 gold-standard scenarios**, cutting hallucinations from **13.0% to 3.0%**.
- Built the **retrieval layer** with **all-mpnet-base-v2** (768-dim) embeddings in a **ChromaDB HNSW cosine index**, attaching metadata (category, severity, tags, confidence) for **filtered top-K retrieval** and a **70% similarity relevance gate**.
- Engineered **prompt and context assembly** grounding **Llama 3.1 8B (Ollama)** in retrieved scenarios, tuning temperature, token budgets, and context windows for **960 ms** text and **1.3 s** voice latency.
- **Led the validation layer for both platforms**: **RAG guardrails** with a NENA-based **ontology validator**, **LLM** and **NLI-based hallucination checks**, PASS/REVIEW/FAIL scoring, and an **85% similarity** gate.
- Ran **RAG ablation studies** showing retrieval drives quality (**F1 0.81 vs. 0.59** and **97.6% vs. 49.7%** pass rate without it), while ontology validation acts as a safety guardrail that cuts hallucinations from **9.5% to 3.0%**.
- Built a **RAG research assistant** with **LlamaIndex, pgvector,** and **FAISS**, raising **recall@10 from 61% to 88%** (evaluated with **RAGAS**) and serving **40+ researchers** at **190 ms p95** latency.
- Built real-time **trace-log analytics** (event streams, Task Completion Rate, Average Field Latency, goal-alignment scoring) feeding an instructor **performance dashboard** for adaptive feedback.

- Applied **graph-based analytics** and **ensemble models** to large-scale disaster datasets on the **ORC Hopper HPC cluster**, forecasting community coping trends with an **RMSE of 0.89**.
- Implemented **model explainability** using **SHAP** for **intent classification** and **behavioral anomaly detection** models, guiding threshold tuning and model governance; the **LightGBM** anomaly detector reduced false positives by **25%**.
- Fine-tuned **Llama 3** with **LoRA/PEFT** and **QAT**, cutting model size by **40%** within **3%** of full-precision accuracy.
- Took models and RAG services from **proof of concept to production** with **MLflow, FastAPI, Docker,** and **CI/CD**, cutting release cycles from **2 weeks to 5 days** with **A/B testing**.

Carelon Global Solutions (Elevance Health) | ***Machine Learning Engineer*** **Bengaluru, India | May 2022 - August 2023**

- Built NLP models using **BERT** and **BioBERT** for clinical entity extraction from unstructured medical text, achieving **80% accuracy** and reducing processing time by **55%**.
- Improved **Scikit-learn** and **XGBoost** models through feature engineering and hyperparameter tuning, increasing accuracy by **8%** and cutting training time by **15%**.
- Containerized ML inference services with **Docker, Kubernetes,** and **Jenkins CI/CD**, improving system uptime by **20%** and enabling real-time EHR integration via **REST APIs**.
- Engineered ML-ready ingestion of **400K+ XML/JSON healthcare records** into **Snowflake** and **Databricks**, accelerating model training by **60%** and reducing preprocessing time by **70%**.
- Built a daily **NPPES** ingestion workflow that improved ML data accuracy by **74%** and cut processing overhead by **60%**, and built **PySpark** ETL on **AWS (Glue, SageMaker)** with **Delta Lake**, improving data reliability by **30%** under **HIPAA**.
- Designed **20+ enterprise ETL pipelines** orchestrated with **Apache Airflow** and **AWS (S3, Lambda, Athena)**, boosting throughput by **70%** and achieving **99.9% pipeline reliability** with end-to-end lineage tracking.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - April 2022**

- Aligned ETL pipelines with business KPIs on the **Seven Plus Locations** project, improving claims data accuracy.
- Redesigned **claims-balancing ETL workflows** and automated **medical invoice processing** pipelines, reducing operational overhead by **60%** and manual errors by **70%**.
- Delivered self-service BI pipelines with **Tableau, Power BI,** and SQL-based models, reducing reporting latency by **40%** and enabling anomaly detection for healthcare analytics.
- Migrated large-scale processing workloads from **Hive** to **Apache Spark/PySpark**, and moved **Sqoop/Hadoop** ingestion to **Azure Data Services**, improving performance, scalability, and pipeline efficiency.
- Developed **RESTful microservices** with **OAuth 2.0** and API versioning, and designed **MongoDB** data models with **Python/Java** validation scripts (UDFs) to improve data quality and consistency.

PROJECTS

Agentic AI-Based Ticket Automation System | SecureGPT, LLaMA, AWS SQS, JIRA, NIST-CSF

- Built an **agentic AI** pipeline with **SecureGPT** and **LLaMA** on **AWS SQS** event queues that monitors cybersecurity threats and auto-generates **JIRA** tickets under **NIST-CSF**, reaching **93% automation accuracy** and cutting triage time by **70%**.
- Used **LLM prompt engineering** to classify and rank events, improving scalability by **50%** and cutting operational risk by **40%**.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, K. B. Laskey, L. Wells II, H. Purohit. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." *ACM WSDM '26*. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, and H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." In *IEEE PST Conference*, 2026.